



Team GSRacers Design Report 2026

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Abstract

The BAJA SAE, the ATV design competition offers undergraduate students a platform to use engineering science principles and demonstrate their expertise in the automotive industry.

The objective of the team is to design and manufacture a robust, reliable, and commercially viable off-road electric vehicle. This design report presents the key design philosophies and technical decisions incorporated into our vehicle which will compete in e-BAJA '26.

The team has been actively participating in BAJA competitions since 2007, with prior experience in m-BAJA events and 2 previous participations in e-BAJA. Entering our 3rd e-BAJA event, the focus this year has been on further strengthening our expertise in electric vehicle systems, electronics, and integrated vehicle design. A significant improvement in the design methodology was achieved by prioritizing performance, ergonomics, and system integration from the initial stages. This approach has enabled early identification and resolution of potential issues, thereby improving overall vehicle reliability ahead of the competition.

In addition to performance, a key secondary objective is to ensure that the vehicle successfully completes the full endurance event. This has driven increased emphasis on the durability and reliability of critical subsystems, particularly the suspension and drivetrain. The purpose of this design report is to document the systematic design process, engineering methods, and decision-making strategies adopted in the development of the vehicle.

Introduction

The BAJA SAE competition challenges undergraduate engineering students to design, build, and race off-road vehicles capable of conquering rugged terrain. By immersing themselves in automotive design literature, our team has developed the expertise to create a competitive vehicle. Drawing from past experiences and industry insights, we have prioritized key design elements to ensure our vehicle's performance and safety. These elements include universal ergonomics, modular components, and robust driver rollover protection. Through this comprehensive approach, we aim to demonstrate our engineering progress and showcase our vehicle's capabilities in the demanding BAJA SAE competition.

SPECIFICATIONS	VEHICLE 2025
Vehicle Dimensions(L*B*H) (inch)	76*57*61
Kerb Weight	200Kg
Wheel Base	53"
Track Width	50"/50"
Ground Clearance	13"
Height of CG from ground	22.5"
Tyre Size (Front/Rear) (inch)	23*7-10
Frame type and Material	Tubular Space Frame/AISI4130
Suspension Type (Front /Rear)	Parallel SLA Suspension
Springs and Dampers	Afco Racing 63 series Shock + Coil Over Dual Stage Spring Rate
Brake Type (Front/Rear)	Disc Brake
Steering Type	Centre tapped rack and Pinion
Static Steer Ratio	44.44
Transmission	Chain and Sprocket
FAW to RAW Ratio	32:68
Maximum Acceleration	4.65 m/sec ²
Maximum Speed	51.98 km/h
Gradeability	83%
Stopping Distance	7.23m

ROLLCAGE

AIM AND OBJECTIVES

1. Driver comfort and safety.
2. Achieve the best strength-to-weight ratio possible.
3. Install subsystems and adhere to reset requirements.
4. For easy serviceability of other components.

DESIGN CONSIDERATIONS

1. The side impact Member is redesigned with a bent to increase front compartment volume and provide better driver ergonomics.
2. The distance between FBM Members has been increased to 518 mm to increase the view sight of the driver.
3. The rear compartment has been meticulously configured in accordance with the transmission and CAE analysis outcomes.

BENT MEMBERS

1. Bent members also contribute to welding reduction, thereby aiding in the overall weight reduction of a structure, which came out to be 24.3 kg and cost of welding is also reduced by implementing more bent members.
2. The complexity of the structure has also been reduced by employing bent members rather than two welded members and the structural symmetry has been enhanced.

TESTING AND VALIDATION

1. The Driver sitting space was then optimized to provide anthropometric comfort to the driver.
2. The foam and paddings are provided at hot points for driver comfort and sharp edges are removed.
3. A rollcage prototype was created to check all the clearance and comfort.
4. The design of the front compartment prioritizes driver comfort by enhancing spaciousness and aligning with front transmission requirements. The separation between the front transmission and the cockpit measures 7 inches, providing ample room for the front transmission assembly.
5. The cockpit is designed with driver comfort in mind, taking into consideration all clearances from the roof and sides, steering clearance of 8.67 inches, and ample legroom.
6. The VIEW ANGLE is maintained by inclining the SIM with a horizontal plane with 3 degrees.
7. In order to distribute the front impact load and increase overall strength, Gussets are provided between SIM and FBM also triangulation is used to distribute load and increase strength.

DRIVER COMFORT AND ERGONOMICS

AIMS AND OBJECTIVES

1. Enhance overall driving experience by avoiding driver fatigue.
2. The vehicle design prioritizes ergonomics for the driver, ensuring safe driving by keeping physical efforts within recommended limits.

1. Cockpit Length: One of our main concerns throughout the design of the Roll-Cage and the arrangement of various equipment in the driver's cabin was driver ergonomics. The cabin has been designed to be secure, comfortable and this time, we have designed the cabin by using computed values to minimize the cockpit's over spacing while ensuring that all of the rules from the rulebook were followed.

2. Side Impact Member: To increase the driver's field of view, SIM has been given a slight inclination from the horizontal plane. The bend from the SIM members has been removed for greater ankle/feet mobility.

3. Rear Roll Hoop: It is inclined at an angle of 7° to reduce the stress developed on the driver's spinal cord and increase the rear compartment spacing.

4. Seat: Suspension seat i.e. sling seat is mounted this year as this design of seat takes the shape of driver's lower body and also keep the driver isolated from chassis vibrations. this year H point is above the foot plane for the

5. Safety equipment: Five-point seat belt, bumper, motocross helmets, tear off goggles, emergency push-to off kill switches, GPS device to track vehicle location and specially fire resistive vinyl covering the firewall sheet and aluminium sheet for paneling are used to ensure the safety of the driver.

6. Damper Tuning: Shock absorbers are tuned and springs are

selected as per the track in the event to ensure better performance and increased driver comfort.

- The basic requirement of driver comfort is to lower down vibrations reaching to the driver. It requires avoiding the resonance of sprung mass with every excitation coming on it. The road excitation is in the range of 10 Hz while the motor is in the range of around 60 Hz. So, we decided to target the natural frequency for roll cage greater than 60Hz for motor excitations. The modal analysis & the frequency response analysis with excitation from motors and road, and measurements at driver hot points gave an approximate prediction of comfort. The Driver sitting space was then optimized to provide anthropometric comfort to the driver. The foam and paddings are provided at hot points for driver comfort and sharp edges are removed avoiding resonance of sprung mass with every excitation coming on it.

DRIVER'S ERGONOMICS PARAMETERS		
Parameter	Requested Value	Design Value
Elbow Angle (in deg.)	125-140	130°
Knee Angle (in deg.)	120-150	125°
Angle at Back (in deg.)	8-150	13°
Seat Angle (in deg.)	90-120	100°
Steering wheel angle (in deg)	30-50	45°
Foot Angle (in deg.)	70-150	95°

Calculations

Second Moment of Inertia (I):

- Rulebook Primary: $D_o = 25.4 \text{ mm}$, $D_i = 19.4 \text{ mm}^4 \Rightarrow I = 13,478.6 \text{ mm}^4$
- Vehicle 2026 Primary: $D_o = 29.21 \text{ mm}$, $D_i = 25.91 \text{ mm} \Rightarrow I = 13,612.4 \text{ mm}^4$
- Rulebook Secondary: $D_o = 25.4 \text{ mm}$, $D_i = 19.4 \text{ mm} \Rightarrow I = 13,478.6 \text{ mm}^4$
- Vehicle 2026 Secondary: $D_o = 25.40 \text{ mm}$, $D_i = 23.40 \text{ mm} \Rightarrow I = 5,714.23 \text{ mm}^4$

Bending Strength (Sb):

- Rulebook Primary: $S_b = 365 \times 10^6 \times 13,478.6 \times 10^{-12} / 12.7 \times 10^{-3} = 387.37 \text{ N-m}$
- Vehicle 2026 Primary: $S_b = 460 \times 10^6 \times 13,612.4 \times 10^{-12} / 14.6 \times 10^{-3} = 428.88 \text{ N-m}$
- Rulebook Secondary: $S_b = 365 \times 10^6 \times 5,152.2 \times 10^{-12} / 12.7 \times 10^{-3} = 148.09 \text{ N-m}$
- Vehicle 2026 Secondary: $S_b = 677 \times 10^6 \times 5,714.23 \times 10^{-12} / 12.7 \times 10^{-3} = 304.60 \text{ N-m}$

Bending Stiffness ($K_b = E \times I$):

- Rulebook Primary: $K_b = 13,478.6 \times 10^{-12} \times 205 \times 10^9 = 2763.13 \text{ N-m}^2$
- Vehicle 2026 Primary: $K_b = 13,612.4 \times 10^{-12} \times 205 \times 10^9 = 2790.54 \text{ N-m}^2$
- Rulebook Secondary: $K_b = 5,152.9 \times 10^{-12} \times 205 \times 10^9 = 1056.34 \text{ N-m}^2$

Vehicle 2026 Secondary: $K_b = 5,714.23 \times 10^{-12} \times 205 \times 10^9 = 1171.41 \text{ N-m}^2$

POWERTRAIN

AIM AND OBJECTIVE

Aim & Objectives:

1. Ensure the motor operates at peak power and transmits power in the most efficient manner for maximum performance.
2. Maintain the vehicle's weight distribution and centre of gravity within acceptable limits.
3. Maximize overall vehicle range
4. Achieve high acceleration and torque output for rough terrain while ensuring high-speed capability.

DESIGN CONSIDERATION

FAILURES:

1. Improper fuse rating selection (48V, 150A required) led to recurring fuse blowouts under normal load conditions.
2. The RTD circuit had design deficiencies, causing the control logic to malfunction
3. Disengagement of CV Shaft.

CHANGES:

1. Circuit breaker of 48V, 150A is selected, having manual isolation & automatic over current protection.
2. A dedicated RTD sequence is designed using STM8 microcontroller & implemented on custom PCB.
3. The diameter of the propeller shaft has been modified from 20 mm to 17 mm.

MATERIALS SELECTION

The materials used in the design of the system by taking in consideration the reduction of rotational moment of inertia as well as maintain the compliance with design and working characteristics. Material selected to ensure performance, durability, safety, and cost-efficiency.

Materials used in design of the system includes:

20MnCr5:

Used in the manufacturing of the gears, gearbox shafts, propeller shaft and differential gears to withstand line contact and fatigue stresses.

Al7075-T6

It was employed to provide the required strength of U-Joints, Rear Drive Shafts and it has high strength to weight ratio and corrosion resistance.

Al 6061-T6:

Differential and Gearbox Casings are manufactured using the material Al6061-T6

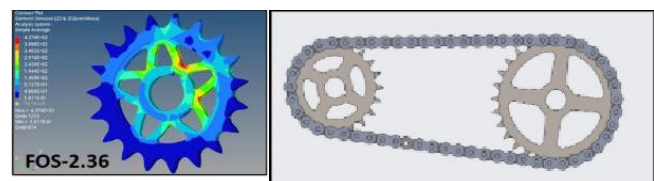
DRIVETRAIN CALCULATION

Dynamic radius	0.28 m
Aero drag	58.65
Rolling resistance	134.88
Max. Grade angle	40
Grade resistance	1734.08 N
Total tractive force	1927.62 N
Total reduction required	9.132
Wheel velocity	51.98 m/s
Grade %	83 %
Acceleration	4.65 m/s ²
Total driveline reduction	9.132: 1
Gearbox reduction	7.58: 1
Required chain drive reduction	9.132/7.58=1.2
Drive sprocket teeth (z1)	19
Driven sprocket teeth (z2)	1.2*19≈ 23
Chain links (Ln)	54
Chain length	685.8 mm

Chain drive calculation:

Parameter	Primary sprocket	Secondary Sprocket
PCD	77.16 mm	101.3
No of teeth	19	23
Maximum Torque	57 mm	75 mm
Face Width	7.2 mm	7.2 mm

Chain Type	8B
C to C	200 mm
Chain Length	685.8 mm
Pitch	12.7 mm
Links	54



PROPELLER SHAFT:

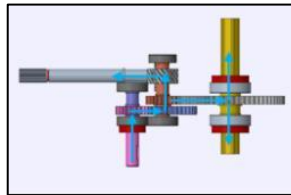
The primary objective of propeller shaft is to transmit power from the gearbox to the differential efficiently. To transmit power, the propeller shaft utilizes a rigid muff coupling at one end, while the opposite end is directly inserted into the differential to prevent misalignment caused by excessive stresses due to bending of the propeller shaft at bump and droop.

Diameter	17 mm
Type	Solid shaft
Length	850 mm
Material	20MnCr5

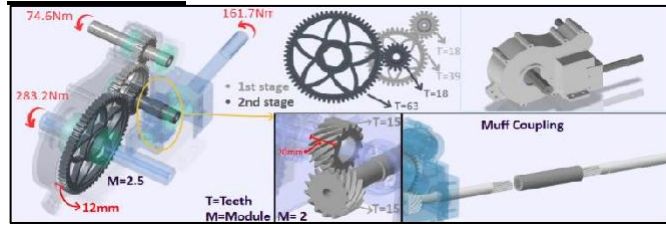


GEARBOX:

- The major objective of gearbox is to ensure smooth power transfer from the motor to the wheels, with optimal performance, durability, and efficiency.
- The major aim in developing the gearbox was to obtain exact gear ratios of 2.16 in first stage and 3.5 in the second stage.
- It also aids in achieving appropriate torque in both the front and rear wheels.



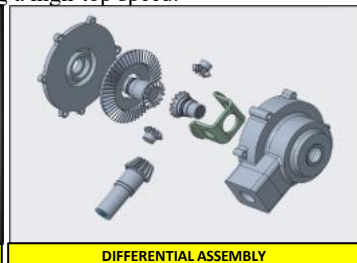
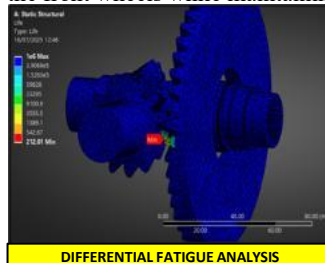
Gear Factors



DIFFERENTIAL:

The open differential with reduction of 3.5 is used to distribute Power laterally between the front tires.

The primary goal in designing the differential is to get adequate torque on the front wheels while maintaining a high-top speed.

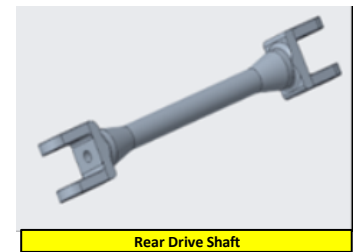


DIFFERENTIAL (GEAR FACTORS):

Parameter	Pinion	Crown gear	Spider gear	Side gear
Module	3	3	3	3
Pressure angle	20	20	20	20
No. of teeth	14	49	10	15
Torque max	102.8	359.15	359.15	359.15
Torque min	21.85	76.46	76.46	76.46

FINAL DRIVE FOR REAR WHEELS:

- U-Joints were the best option for power transmission from gearbox output to stub axle due to its compliance with rear suspension geometry, less complexity, and more reliability.
- Integrated hollow drive shafts are used which optimizes rotational inertia.



Diameter	25mm
Type	Hollow shaft
Length	409mm
Material	Al7075 T6

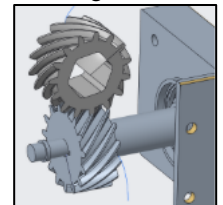
FOR FRONT WHEELS:



- CV Rzeppa joints at both the ends were OEM with 3-inch telescopic plunges.
- The shaft was designed in order to reduce weight and impart strength.
- It has articulation of 48 degrees.

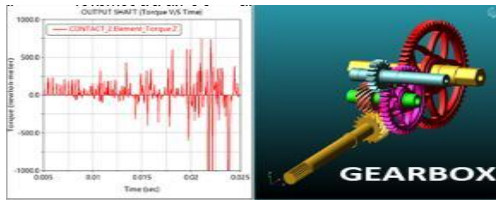
CROSS HELICAL GEARS

- We selected a cross-helical gear system with a 1:1 Transmission ratio to transfer power from the gearbox to the front differential, replacing the conventional Bevel gear setup.
- This choice allowed us to lower the vehicle's center of gravity by approximately 1.8 inches, improving overall stability.
- This arrangement provides smoother gear engagement, leading to reduced impact loads and lower levels of vibration during operation.



ANALYSIS AND OPTIMIZATION

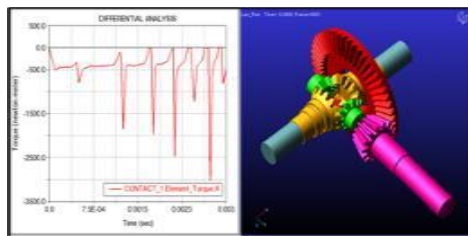
1. Gearbox Analysis on Adams



Contact Torque on input shaft vs output shaft

Min	Max	Avg	RMS
-1103.04	756.44	35.221	77.308

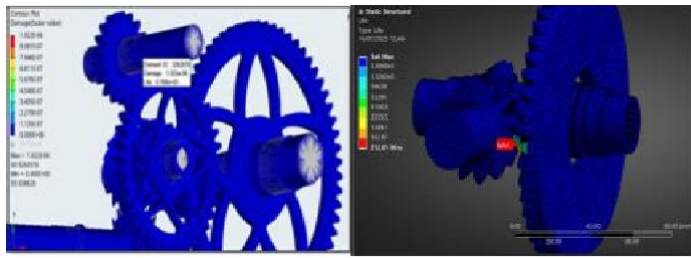
2. Differential Analysis on Adams



Contact Torque on Pinion V/S Crown Gear

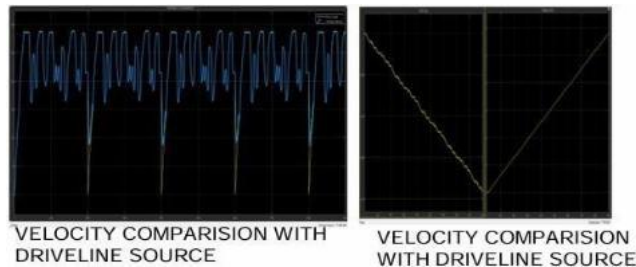
Min	Max	Avg	RMS
-3029.948	2.857E-1	-457.175	689.781

3. Fatigue Analysis of gearbox and differential



TESTING AND VALIDATION

Full vehicle validation on MATLAB



HIGH VOLTAGE SYSTEM

BATTERY

The tractive system uses 21700 NMC cells, each rated at 4.5Ah. The peak configuration is 13S-17P, giving a 48V

Nominal pack with a total capacity of 80Ah. This Arrangement provides high energy density, stable thermal Behavior, and sufficient discharge capability for peak motor Loads. NMC cells were selected due to their high gravimetric Energy density, stable thermal characteristics, and superior Cycle performance under high discharge loads.

BATTERY CALCULATIONS

Actual Capacity	$= (V \cdot AH \cdot \text{efficiency} \cdot \text{Peukert's effects}) / 1000$ $= (48 \cdot 80 \cdot 0.98 \cdot 1) / 1000$ $= 3.7632 \text{ Kw}$
Battery Consumption	$= (V \cdot I) / \text{SPEED}$ $= (48 \cdot 70) / 51.98$ $= 64.64 \text{ Wh/km}$
Distance Covered	$= (\text{Actual capacity}) / (\text{Battery Consumption})$ $= (3763 \text{ W}) / (64.64 \text{ Wh/Km})$ $= 58.21 \text{ Km}$

BMS

BMS stands for battery management system, our vehicle uses JBD-SP24S004 V1.1 Smart BMS, a programmable system compatible with 14–24 series lithium packs and rated for 40–200A continuous operation.

The BMS was chosen due to its wide current range (suitable for peak traction loads), robust protection logic, and reliable cell-level monitoring.

Key Features & Functions

- ❖ Cell monitoring
- ❖ Current & temperature protection
- ❖ Balancing & communication
- ❖ Control outputs

MOTOR

The vehicle is powered by a **6-phase Permanent Magnet Synchronous Motor (PMSM)** rated at **5 kW (48 V)**.

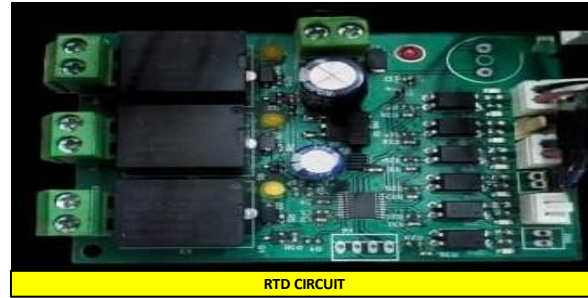
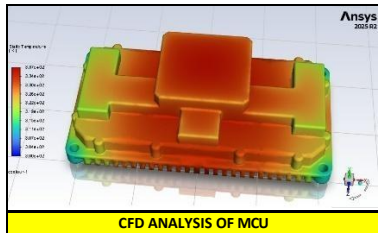
The motor provides a rated torque of 19 Nm and a peak torque of **57 Nm**, enabling strong low-speed traction essential for off-road performance. The peak power output is $\approx 7.2 \text{ kW}$, offering sufficient acceleration and gradeability for dynamic events.

MOTOR CONTROLLER UNIT (MCU)

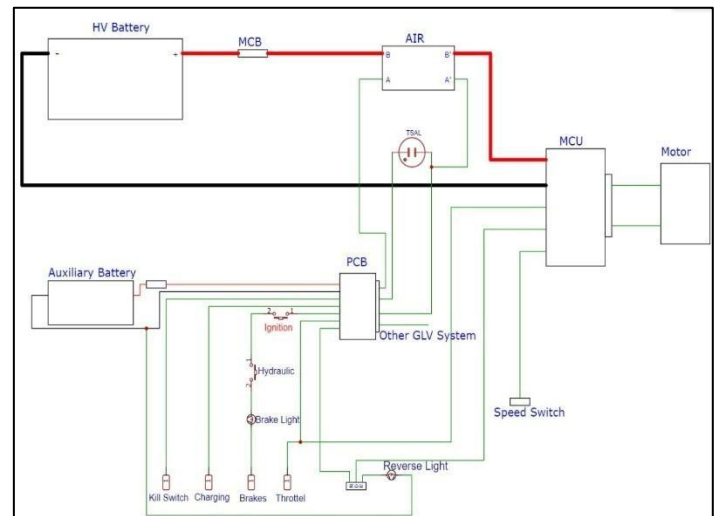
MCU stands for Motor Control Unit. It's a crucial component in electric vehicles that serves as the "brain" of the Electric system. Its primary function is to efficiently manage the flow of electrical energy of battery to electric motor ensuring optimal performance, efficiently and safely.

Key Functions

- ❖ Reads, filters & validates throttle sensor input.
- ❖ Controls **NEUTRAL/ DRIVE/ REVERSE** selection.
- ❖ Implements three selectable power modes.



COMPLETE CIRCUIT DIAGRAM



ACCUMULATOR ISOLATION RELAY

AIR is used as a primary safety component to electrically isolate the high voltage battery from the rest of the vehicle, having current rating of **200A** and actuating voltage of 12V. It plays a critical role in emergency shutdown and pre charge safety.



CIRCUIT BREAKER

Circuit breaker is used as a primary safety component and protection device in high voltage DC battery line. It provides manual isolation and automatic over current protection, ensuring safe operation during faults having rated current of **150A**.

LOW VOLTAGE SYSTEM

Auxiliary battery

The auxiliary battery is 12V, 24Ah battery equipped with an internal battery management system. It independently powers all low-voltage electronics, ensuring that control and safety systems remain active even the high voltage system is isolated.

Ready to drive (RTD) circuit

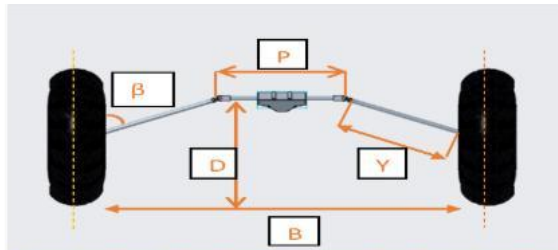
A dedicated RTD circuit is designed using and implemented on a Custom PCB using a STM8 microcontroller. The RTD system Ensures that vehicle enters the drive ready state only after all safety conditions are satisfied. The PCB integrates seamlessly. It activates RTD buzzer before enabling traction system.

STEERING

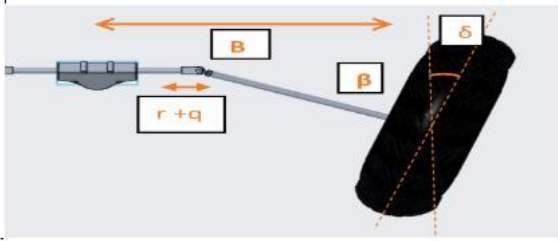
AIM AND OBJECTIVES

1. To minimize turning radius
2. Reduce steering effort while minimizing response time for quicker handling.
3. Minimize bump steer to ensure directional stability of the vehicle.
4. Ensure precise directional control across various terrains.

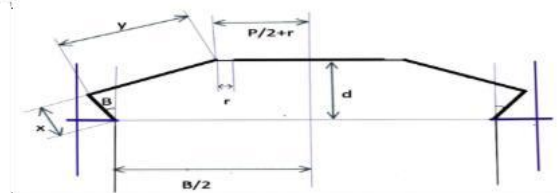
DESIGN CONSIDERATIONS



$$Y^2 = [B/2 - (p+2r)/2 + x \sin \beta]^2 + [\delta - x \cos \beta]^2$$



$$Y^2 = [B/2 - (p/2 + r + q) + x \sin(\delta + \beta)]^2 + [\delta - x \cos(\delta + \beta)]^2$$

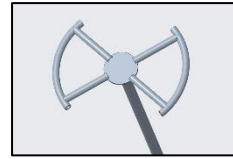


$$Y^2 = [B/2 - (p/2 + r + q) - x \sin(\delta + \beta)]^2 + [\delta - x \cos(\delta + \beta)]^2$$

In addressing the objectives, the steering system design was approached using three key equations involving four variables: rack length, knuckle length, knuckle angle, and rack travel. The steering axis along with the knuckle was also shifted altogether. Certain constraints, predefined by other departments, influenced the design, including the rack position relative to the wheel axis, rack height, and tie rod length. These factors were carefully integrated into the design to meet performance and handling targets effectively.

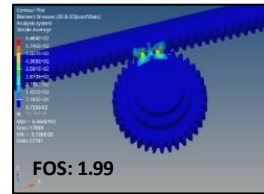
Tractive force	404.25 N
Lateral moment	4246.6 N-m
Tractive moment	13.13 N-m
Lateral moment	13.01 N-m
Total moment	26.14 N-m
Tie rod force	301.30 N
Rack force	253.69 N
Pinion Moment	5.68 N-m
Steering Effort	74.5 N

1. Steering column length and angle designed for driver comfort and to achieve the clearance from the driver.
2. Rack placed 5 inches from the front wheel axis, based on best iterated results in MSC Adams.
3. Grooves in the casing with clearance for placing linear slider bearing to reduce play in rack and casing.



4. Steering wheel made from curved tubes to better place the wheel according to the hands of the driver.

ANALYSIS AND OPTIMISATION

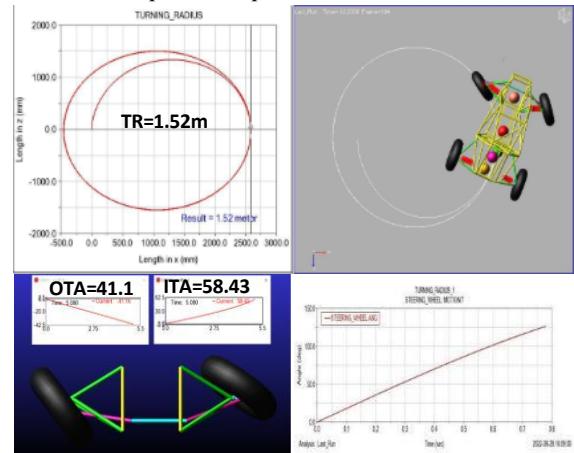


1) Finite Element Analysis (FEA)

Utilized FEA software to analyze components like steering column, rack, and tie rods for strength and durability.

2) Dynamics Simulation -

Implemented multibody dynamics simulations on MSC ADAMS 2024 to analyze steering behavior during sharp turns, emergency maneuvers, and rough terrains and calculate bump and droop.



Optimization Strategies –

Material Optimization: Enhance strength-to-weight ratio and durability of steering components.

Geometry Optimization: Adjusted rack length, knuckle geometry, and steering column angles through iterative simulations to achieve desired performance metrics.

Ergonomic Optimization: Analyzed steering column length and angle for optimal driver ergonomics, ensuring a balance between comfort and control.

MATERIAL SELECTION		
Components	Properties	Materials
Rack and Pinion Gear	High hardness, good toughness	20MnCr5
Rack and Pinion casing	High Strength-to-weight ratio, corrosion resistance	Al6061-T6
Column, Tie Rod	High bending and tensile Strength	AISI-4130

PARAMETERS	FRONT		REAR
WHEEL TRAVEL (inch)	BUMP	6	7
	DROOP	4	3
CAMBER (deg.)	KERBW T.	0	0
	IMPACT	-.65/ -.195	-1.54/ 0.94
CASTOR (deg.)	4		0
KPI (deg.)	8		0
SCRUB RADIUS (mm)	35.148		0
TOE CHANGES	Static	0	0
	Dynamic	-4.7	-1.39
MOTION RATIO	0.767		0.73

SUSPENSION

AIMS AND OBJECTIVES

1. To maintain wheel contact with the ground.
2. To achieve desired Roll Stiffness and Camber characteristics while Cornering.
3. To protect the rollcage from Bumps and Vibrations.
4. to ensure ride comfort and safety of driver.

DESIGN CONSIDERATIONS

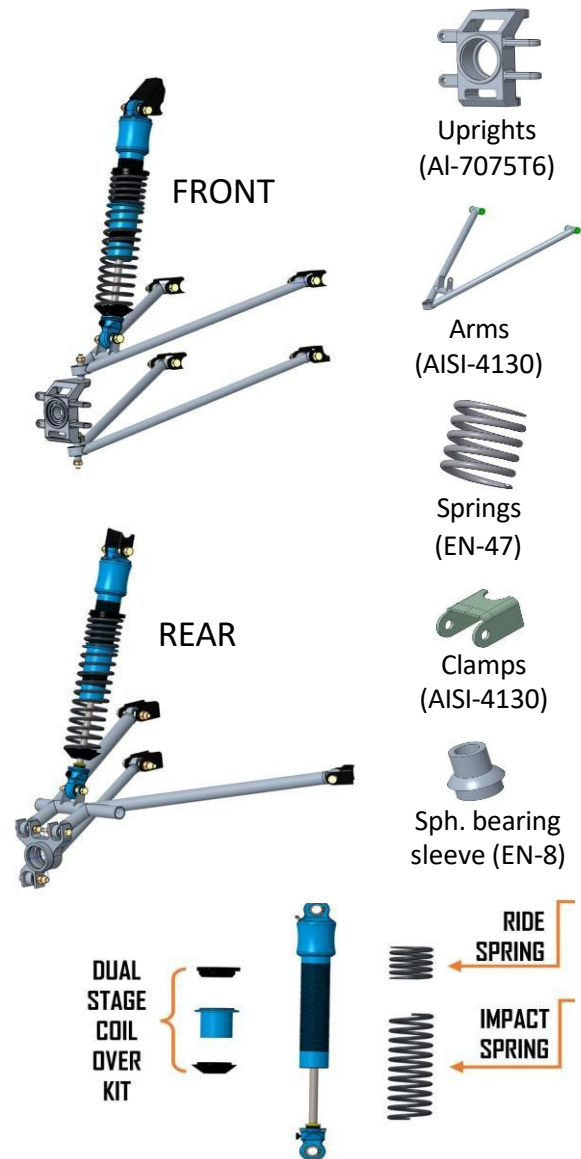
Failures-

1. **UBJ Bearing Failure:** Caused by the damper's inability to absorb sufficient forces.
2. **Spring Stiffness Issues:** Excessively stiff springs reduced ride comfort, harmed overall performance, and increased the risk of UBJ bearing and CV shaft failure.

GEOMETRY

Front Suspension: The SLA suspension with parallel links delivers strong stability and traction by maintaining a consistent contact patch and preventing tire slip. Its geometry provides effective camber control and minimizes bump steer for predictable handling. The front suspension offers 6 inches of bump and 4 inches of droop travel to maximize grip and improve overall performance.

Rear Suspension: A modified H-arm rear suspension with a parallel camber link is used. Heim joints are replaced with spherical bearings for better durability and strength. The H-arm sits above the wheel center to avoid interference and improve damper mounting, giving a favorable motion ratio. The system provides 7 inches of bump and 3 inches of droop to maintain traction and contact patch.



AFCD 63 SERIES 7"-3" SHOCK

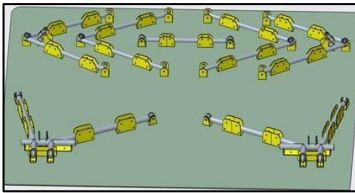
Material Selection		
Arms	High bending and tensile strength	AISI-4130
Springs	High fatigue resistance, good elasticity	EN-47
Upright, Hub, Flange	High strength-to-weight ratio, good machinability	AL-7075 T6
Stub	High toughness, high tensile strength, wear resistance	EN-24

COMPONENT MODELLING

The suspension frequency was selected based on ride-comfort limits, which defined the required spring stiffness. Dual-stage springs were used to provide variable stiffness—soft for normal driving comfort and stiff under harsh conditions to prevent bottoming out.

The uprights and A-arms were modeled to match the suspension geometry and analyzed in Hyper Mesh for structural and assembly performance.

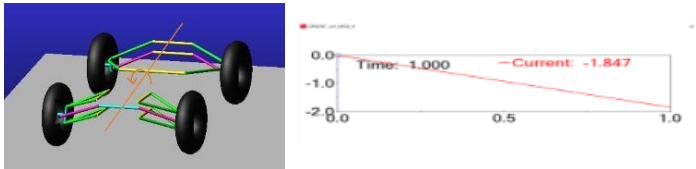
Tube diameters and cross-sections were optimized to balance strength, weight, and manufacturing feasibility.



Additionally, jigs and fixtures were specifically designed to ensure precision in the assembly of suspension control arms, reducing the likelihood

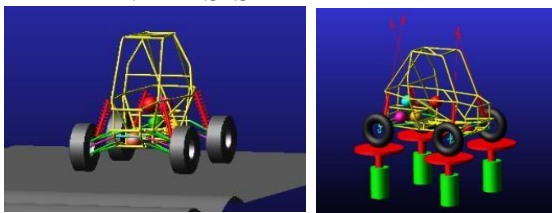
of manufacturing defects and ensuring proper alignment during production.

ANALYSIS AND OPTIMISATION:- ROLL CAMBER CHANGE



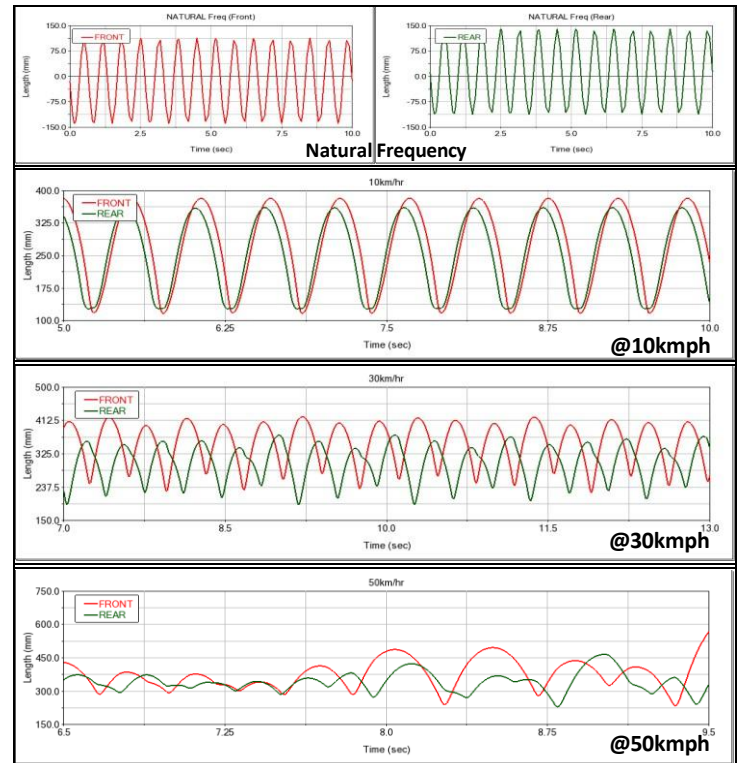
Shown this is the graph of Roll Camber Kinematic Analysis performed on the vehicle's front suspension geometry. It tells us about the camber change that will occur if the vehicle undergoes cornering and a prediction for the contact patch area.

RIDE ANALYSIS



METHODOLOGY:-

- 1) Four Post Testing Rig (for natural frequency) & vehicle on sinusoidal path (for ride frequency) on MSC Adams 2024.
- 2) For the ride analysis, realistic tyres and road model were used instead of the simplified toruses typically used in other analyses. The tyre and road file was updated to match the specifications of the actual tyre.



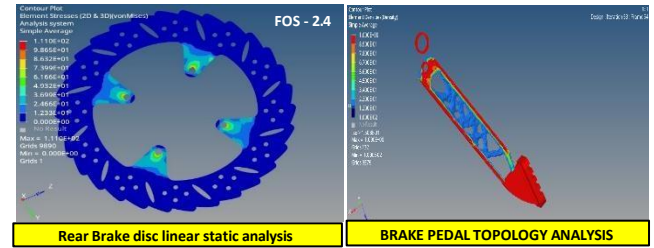
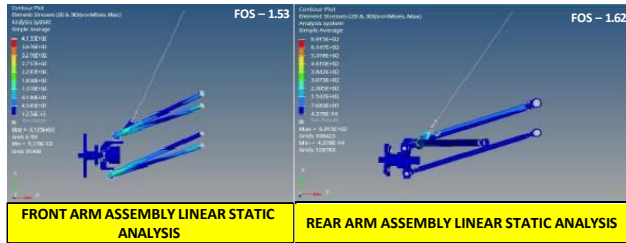
Ride Summary		
	Front	Rear
Ride Rate	6.39 N/mm	11.72N/mm
Spring Rate	12.39 N/mm	26.59 N/mm
Wheel Rate	7.16 N/mm	14.56 N/mm
Frequencies		
Ride Frequencies	Front	1.9 Hz
	Rear	2.1 Hz

Frequency	Natural	Ride @10kmph	Ride @30kmph	Ride @50kmph
Front	1.428Hz	1.846Hz	5.95Hz	9.02Hz
Rear	1.538Hz	1.889Hz	6.451Hz	9.52Hz

ASSEMBLY ANALYSIS

Linear Static Analysis is done to ensure the safety of the suspension assembly.

RESULTS –



BRAKES AND WHEEL ASSEMBLY

AIM AND OBJECTIVES:

- To decelerate the vehicle within the specified stopping distance(8m).
- To ensure the structural integrity and proper functioning of the wheel assembly.

DESIGN CONSIDERATIONS AND SUGGESTED CHANGES:

FAILURE:

- Last year, rear brake disc was overheated which reduces the braking efficiency.
- Last year, thickness of rear caliper mount was 4mm Which then got bent and unintended caliper-disc contact was observed.

CHANGES:

- Rear Brake disc's material is now changed from SS420 to Cast Iron for better heat dissipation.
- Rear Caliper Mount thickness is now increased to 6mm with additional supports for high bending stiffness.

MATERIAL SELECTION:

Component	Material used	Criteria for selection
Front brake disc	SS420	High Wear and Corrosion Resistance
Rear Brake Disc	Cast Iron	Heat Dissipation ,
Brake Pedal	AL7075-T6	Strength-to-Weight Ratio, Fatigue Resistance
Rear Caliper Mount	EN-8	Strength & Toughness, Machinability
Brake Fluid	DOT-3	More Hygroscopic, easy Air Bubble Determination, Cost Effective

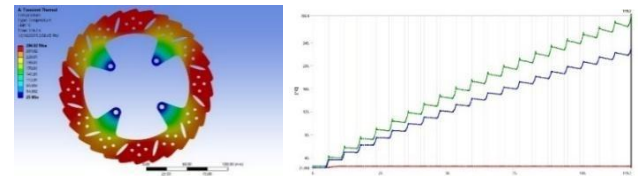
OEM SELECTION:

All rotors will be fitted with KBX fixed calipers, employing a piston diameter of 31.75 mm and maximizing the swept-pad area for superior braking performance. Featuring stainless-steel braided brake lines for effective pressure retention and better braking response, paired with a Bosch master cylinder (15.89 mm bore) to deliver consistent hydraulic force — this setup elevates brake feel and reliability under performance use.

BRAKE PEDAL	SPECIFICATIONS
Material	AL7075
RATIO	6:1
WEIGHT (gm)	200
UTS (MPa)	550
FATIGUE STRENGTH	160 MPa (5E+08 CYCLES)

BRAKE DISC	SPECIFICATION	
	FRONT	REAR
Material	SS420	Cast Iron
Weight	478.5 gm	540.2 gm
Thermal Conductivity	27 W/mK	53 W/mK
Friction Coefficient	0.35	0.45
Compressive Strength	1600 MPa	1280 MPa

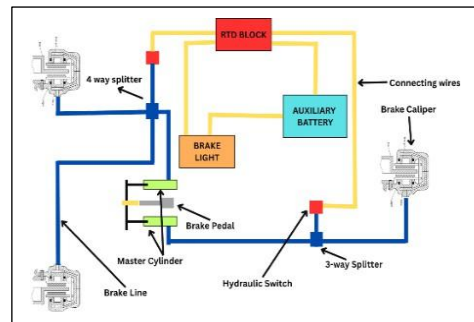
TRANSIENT THERMAL ANALYSIS OF REAR BRAKE DISC:



DISC TEMPERATURE VALIDATION:

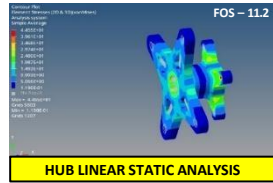
The temperature rise in the disc during braking was validated with the use of a temperature gun. The observed temperature rise of 200 °C was consistent with the CAE result

BRAKE CIRCUIT



HUB:

Using rims with a 4:1 offset and MC with a 15.89 mm bore diameter reduces the size of the wheel hub and bending stress of the hub.



HUB DESIGN BASIS:

1. The hub is designed to reliably handle the bending, shear, and torsional loads that occur during acceleration and braking, ensuring safe operation under all running conditions.
2. The geometry and mounting interface are developed to transmit torque from the shaft to the wheel with minimal losses, improving overall drivetrain efficiency.
3. The hub layout supports smooth and effective braking by maintaining stable wheel behavior and allowing the brake system to operate without unnecessary vibration or play.
4. The design is kept user-friendly, allowing for straightforward manual assembly while ensuring all components seat properly and securely.

MOUNTS:

The master cylinder mounts are now more rigid with additional supports being added to increase overall stiffness. The accelerator and brake pedals are placed on the DLC member. Caliper and disc are mounted to the roll cage this year for better stiffness. The front caliper is attached to the upright, while the front disc is attached to the wheel hub through a countersunk bolt.

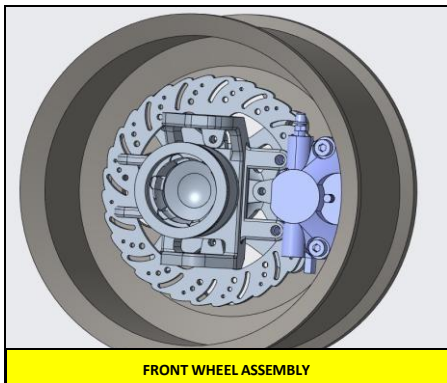
WHEELS & TYRE ASSEMBLY

RIM:

The selection of the rim was the most critical factor, as it needed to accommodate the hub, disc, and calipers. To ensure proper spacing and minimize rotational inertia, the rim's weight and size were carefully considered in collaboration with the design, suspension, and brake departments. The final design utilizes a 4:1 rim offset in the front and a 3:2 rim offset in the rear.

TYRES:

The tyre selection process prioritized torque transmission, top speed, and maximum grip while considering weight reduction for optimal performance. Low Inertia Radial Ply BKT tires, size 23 x 7-10, were chosen to meet these criteria.



Components	Specs
Tyre	23*7-10"
RIM	10*5"

RIM	PCD	OFFSET
FRONT	144	4B+1N
REAR	110	3B+2N

TESTING & VALIDATION

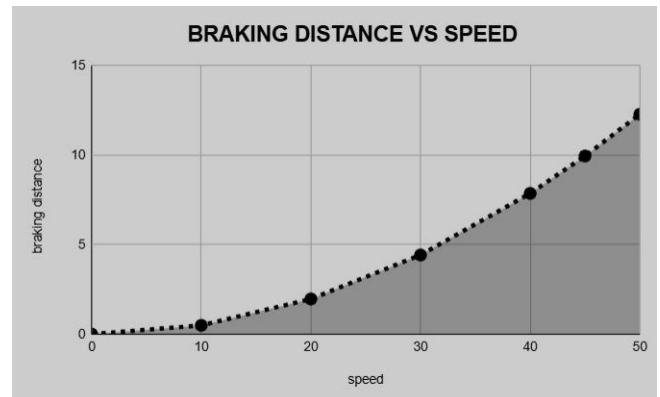
STOPPING DISTANCE:

The brake test was virtually carried out in IPG Car Maker software by accelerating the vehicle at 40 kmph. The stopping distance was found to be 7.88 m.

BRAKE BIAS & PRESSURE:

The adjustable balancing bar is designed to optimize brake force distribution, resulting in simultaneous locking of all four wheels. The rear brakes are assigned 50% of the total force from the master cylinder. Hydraulic pressure gauge is used to verify that the generated pressure matches theoretical calculations and meets the requirements for braking torque.

STOPPING DISTANCE VS SPEED:



CALCULATIONS:

FACTOR	VALUE
Pedal Force	400 N
Force on both MC	2400 N
Pressure in front MC	7.265 Mpa
Clamping Force CF (each side front)	5749 N
Frictional/Braking Force (BF front)	4024.3 N

Braking Torque(front)	314.57 Nm
Pressure in rear MC	4.843 Mpa
Clamping force (rear)	7664.81 N
Braking Force (BF rear)	3449.164 N
Braking Torque(rear)	279.17 Nm

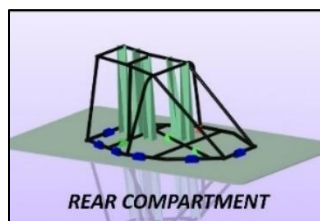
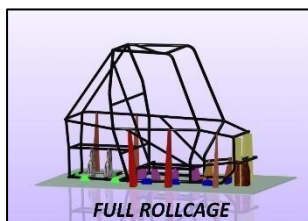
VEHICLE INTEGRATION

AIM AND OBJECTIVES

1. Designing the subsystems to ensure proper interaction between them on assembly level.
2. Keeping the provision for easy and safe maintenance and serviceability, i.e. proper assembly and disassembly
3. To accommodate maximum percentage of targeted users without compromising the comfort of the occupant by following all SAE and ISO standards.

NOTABLE WORKS

1. The suspension links in rear were not in clearance with shaft at full rebound of suspension. Hence it we decided to place Harm above the wheel Centre and mount our suspension on it, as opposed to the conventional design. Further it was also feasible to achieve higher motion ratio in this orientation
2. In order to ensure that the driveshaft is compatible with the suspension and steering subsystems, Rzeppa articulation of 48 deg and shaft plunge of 3.5 inch was provided. To assure the torsional stiffness of the wheel assembly, hexagonal coupling was introduced to the Rzeppa
3. Emergency button helps the driver to send signals to the trouble shooter team in case of an emergency via GPS.
4. Highly stiff flapped Jigs ensured accurate manufacturing of roll cage, further it also ensured proper positioning of hardpoints of suspension and steering. The manufacturing of chassis was conducted in two parts. Firstly, we designed separate jigs and fixture, by taking RRH as our base and assembling the Whole rear compartment on it (like FAB members, RLC etc.) then we assembled it on the master fixture plate to complete the full vehicle chassis. The above manufacturing process avoids propagation of errors due to faulty manufacturing of members and ensures symmetric nature of errors even if they occur.
5. Transmissibility ratio: Flexible mountings with rubber dampers made sure that vibrations induced by the motor and transmission components were properly isolated. The proper and durable assembly of the component to the roll cage was guaranteed by mounts and clamps of the ideal thickness. Below is a graph demonstrating the transmissibility ratio.



OCCUPANT PACKAGING

This year under occupant packaging when have taken into consideration few factors keeping in mind the SAE and ISO standards.

1. Reachability and Operations- In are vehicle column angle is kept 45deg, distance between driver's chest in steering wheel is 245mm and ankle angle for pedal is 95deg at rest (as per the ergonomics point of view).
2. Field View- The field of view of driver sitting inside the cage is roll 135deg*-120deg.
3. For Packing-Panels is open from side, it helps in assembly.
4. Clearance – Knee clearance = 3-inch
Head clearance = 7inch Lateral clearance = 12inch

CONCLUSION

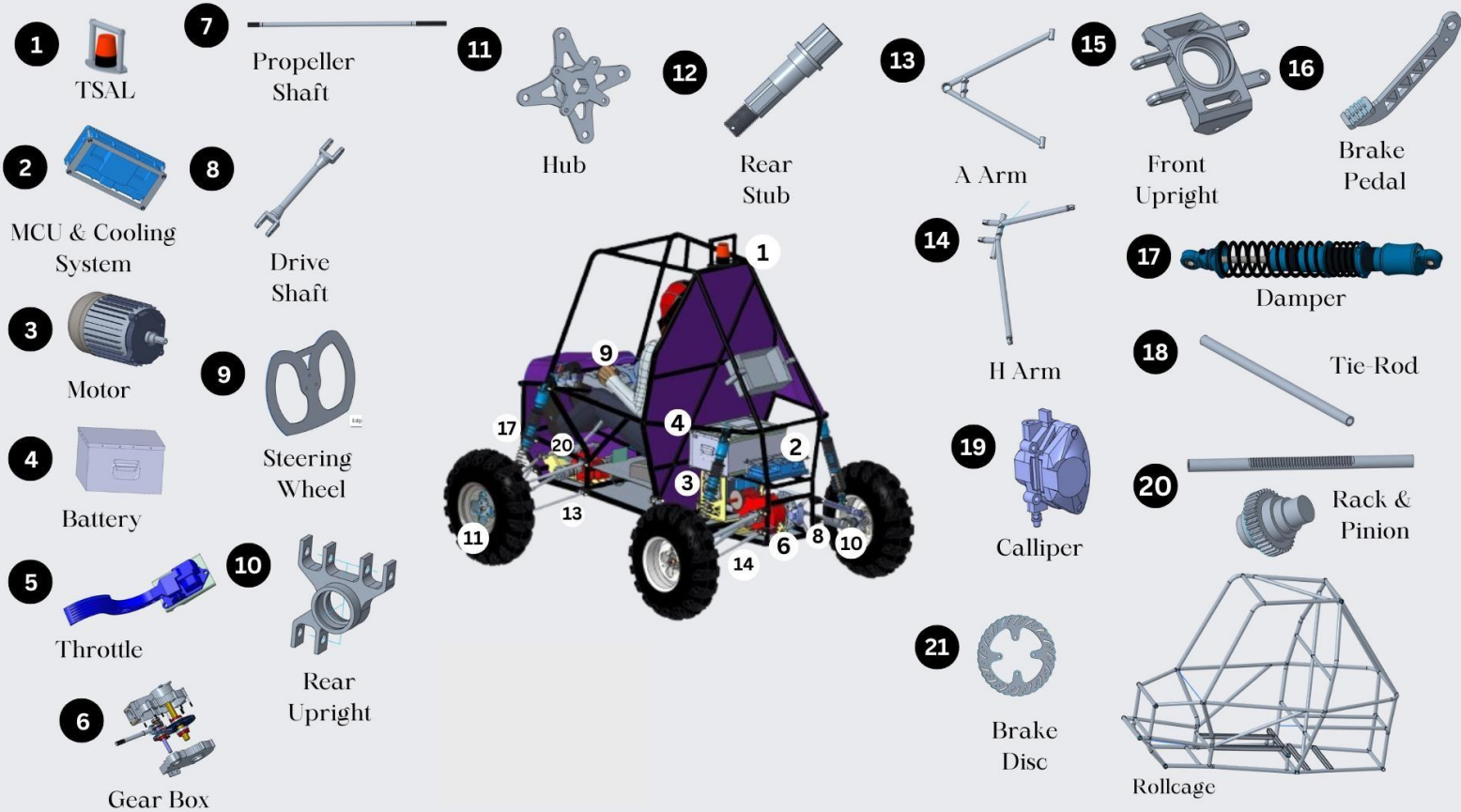
The design methodologies followed this year prominently focused on learning from data and past experiences. Various new design highlights were a result of the experiences we've had with our previous designs and were referenced, and extensive testing was carried out.

REFERENCES

1. Team GS Racers Virtuals Report 2026
2. Team GS Racers design Report 2025
2. Team GS Racers CAE Report 2026
3. <http://www.bajasaeindia.org>
4. Smith, Carroll, Tune to Win, Aero Publishers, Inc., Fallbrook, CA1978.
5. Milliken, William F. & Douglas L., Race Car Vehicle Dynamics, Society of Automotive Engineers, Warrendale, PA 1995.
6. Gillespie, Thomas. D., Fundamentals of Vehicle dynamics
Standard View of Vehicle Assembly.

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E-ATV FULL ASSEMBLY

